

How to Make Meaningful Diagnoses with Abdominal Radiographs: Clues from Natural “Contrast Agents”

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Author affiliations, funding, and conflicts of interest are listed at [the end of this article](#).

See the slide presentation [here](#).

With the rightful ascendance of advanced imaging modalities such as CT, US, and MRI, abdominal radiographs have been relegated to a secondary diagnostic role. Hence, there is a perception of them as quaint relics that offer no useful diagnostic information, deserving only a cursory “nasogastric tube tip in the stomach.”

Yet, as recently as 2021, over 8 million abdominal radiograph studies were performed in the United States alone (Centers for Medicare & Medicaid Services data, courtesy of Jennifer Hemingway and Chi-Mei Liu, Harvey L. Neiman Health Policy Institute). Low cost and universal availability likely play a role, particularly in less affluent health care settings. Consequently, radiologists’ ability to interpret these studies in a manner that adds value to patient care will remain a part of standard clinical practice for years to come. Abdominal radiographs indeed can help to provide valuable clues to diagnosis if the interpreter knows what to look for.

Over the decades, the common appellation “plain film of the abdomen” came to mean a radiograph with no exogenous contrast agents in the gastrointestinal or genitourinary tracts or in blood vessels. But that categorization ignores three intrinsic “contrast agents” available on all abdominal radiographs: gas, calcium, and fat. These intrinsic contrast agents offer insights into varied clinically relevant diagnoses (among them, small bowel obstruction, colonic volvulus, bowel infarction, extraluminal gas, and sites of infection [Fig 1]), and help to identify and characterize calcified (Fig 2) or fat-containing benign and malignant pathologic conditions.

The accompanying slide presentation is designed as a guide for using these “contrast agents” to recognize clinically important findings on abdominal radiographs. Some key points to be gleaned from the presentation include (a) how to develop an orderly approach to analyzing the bowel gas pattern, (b) where to look for pneumoperitoneum when a supine

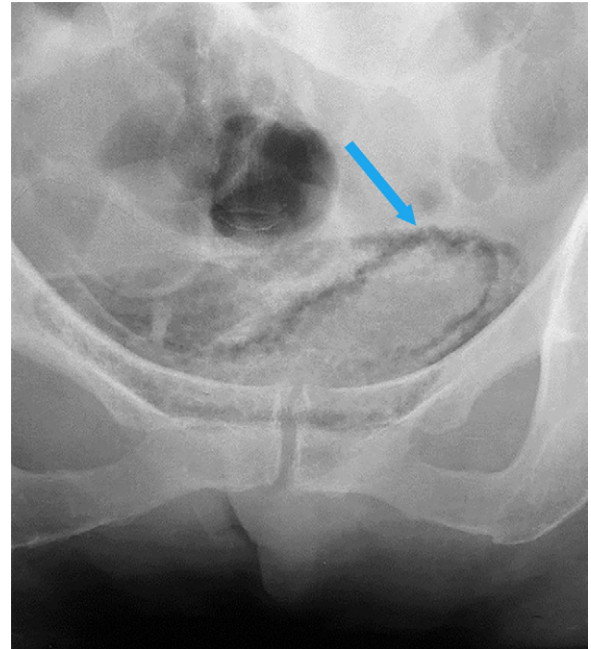


Figure 1. Abdominal radiograph shows emphysematous cystitis, a potentially life-threatening infection, revealed by gas in the bladder wall (arrow).

frontal view is all that you have, and (c) how to use calcifications and fat to find and analyze pathologic conditions.

The goal of the presentation is to provide illustrative diagnostic examples that encourage an orderly and thoughtful approach to interpretation of abdominal radiographs, just as would be expected with a patient’s CT or MRI study. The humble “kidney, ureter, and bladder” (KUB) study still holds important diagnostic clues that it will reveal to the discerning eye.



RadioGraphics 2024; 44(9):e230207
<https://doi.org/10.1148/rg.230207>
 Content Codes: GI, GU

TEACHING POINTS

- Gas, calcium, and fat provide clues to specific diagnoses on abdominal radiographs.
- Clues from gas, for example, can indicate bowel obstruction or perforation.
- Intra-abdominal fat and calcifications can reveal organomegaly, inflammatory processes, and masses.

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Disclosures of conflicts of interest.—**D.A.B.** Paid section editor for UpToDate; honoraria for noon conference lectures and master series from MRI Online (now Medallity); support from the American Roentgen Ray Society for the 2023 Intersociety Meeting; board member of the American Roentgen Ray Society as vice president (2023–2024), president-elect (2024–2025), and president (2025–2026); investor in cultivate(MD). **R.B.D.** Royalties from Elsevier for the book “The Requisites: Genitourinary Imaging.” **J.G.C.** Member of the ACR Radiology Leadership Institute board; stock in companies unrelated to this article. The other author has disclosed no relevant relationships.

Suggested Readings

Baker SR, Cho KC. The Abdominal Plain Film with Correlative Imaging. 2nd ed. Stamford, CT: Appleton and Lange, 1999.
 Ho LM, Paulson EK, Thompson. Pneumatosis intestinalis in the adult: benign to life-threatening causes. *AJR Am J Roentgenol* 2007 Jun;188(6):1604-13.



Figure 2. Abdominal radiograph shows pancreatic serous cystadenoma, identifiable by its stellate calcified central “scar” (oval).

Levine MS, Scheiner JD, Rubesin SE, Laufer I, Herlinger H. Diagnosis of pneumoperitoneum on supine abdominal radiographs. *AJR Am J Roentgenol* 1991;156(4):731-735.

Paulson EK, Thompson WM. Review of small-bowel obstruction: the diagnosis and when to worry. *Radiology* 2015;275(2):332-342.